

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & M_{xy} & 0 \\ M_{xy} & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_v^{(1)}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -M_{xxz} & 0 \\ M_{xxz} & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{\beta(1)}$$

$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{T}_{xy}^{(1)}$$

$$M_{yzx} = \mathbb{T}_v^{(1)}$$

$$\frac{M_{xxz}}{2} = \mathbb{T}_{xy}^{(1)}$$

$$2M_{yzx} = 2\mathbb{T}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ M_{yzy} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_v^{(2)}$$

$$M_{yzy} = \mathbb{T}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & M_{xxxy} \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxzz} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{yzyz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} + 2\mathbb{M}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = -\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & M_{yzzx} & 0 \\ M_{yzzx} & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_v^{(4)}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ M_{yzyz} & M_{yzzx} & 0 \\ -M_{yzzx} & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{yzyz} = -\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxxy} \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzzz} & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyyx} & M_{xyxx} & M_{xyzx} & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$\begin{aligned} M_{xxxx} &= \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\ M_{xxyy} &= -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\ M_{xxzz} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)} \\ M_{xxxxy} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)} \\ M_{xxyx} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)} \\ M_{zzxx} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\ M_{zzxy} &= \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} - 4\mathbb{T}_z^{\beta(1)} \\ M_{yzyz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)} \\ M_{yzzx} &= \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)} \\ M_{yzzz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)} \\ M_{yzzx} &= \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)} \\ M_{xyyx} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{\beta(1)} \\ M_{xyzz} &= \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + 4\mathbb{T}_z^{\beta(1)} \\ M_{xyxy} &= 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & M_{yzzz} & 0 & -M_{yzzz} & M_{yzyz} & 0 \\ M_{xyyx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned} M_{yzyz} &= 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\ M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)} \\ M_{yzzz} &= -2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\ M_{yzzx} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)} \\ M_{xyyx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)} \\ M_{xyxy} &= 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & M_{xxxx} & 0 \\ M_{xxxx} & -M_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & M_{zzxy} & 0 \\ M_{xxyx} & -M_{yyzz} - M_{zzxx} & 0 \\ M_{yzzx} & M_{yzzz} & 0 \\ 0 & 0 & M_{zzxx} \\ -M_{yzzz} - M_{zzxx} & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\
M_{xxxy} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} \\
M_{yyzz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxx} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxy} &= \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} \\
M_{xxyx} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} \\
M_{yzxz} &= \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} \\
M_{yzyy} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{zzxz} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{xyyy} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)}
\end{aligned}$$